SIMATS ENGINEERING

ASSESSMENT TOOL 3: Game Based Learning Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS

COURSE OUTCOME COVERED
COURSE CODE: MLA01

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems.* (BTL 3) Assessment Tool Weightage: 40 %

Total Marks: 50 Time: 60 Mins No. of Questions: 5

Annexure A

Game Based Learning Assessment

Code of Conduct:

I ___Sk. Fawaz Ahamad / 192425390__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
Understanding of the Problem / Game Objective	10%	
Application of AI Search / Problem-Solving Technique	15%	
Successful Completion of the Game / Puzzle	20%	
Efficiency of Solution / Optimal Moves	10%	
Documentation / Evidence of Completion	15%	
Understanding of the Problem / Game Objective	15%	
Originality and Critical Thinking	15%	

Total	100%	

Signature of the Faculty Total Marks Awarded

Answer All the Questions

Q. No.	Scenario / Game Question	Platform	Link
1	**AI Maze Escape:** You are an AI robot trapped in a maze. Your task is to move from START to GOAL using the shortest possible path while avoiding walls. **Challenge:** Find the shortest path and record the number of steps taken.	Scratch	https://scratch.mit .edu
2	**N-Queens Challenge:** You are an AI chess strategist. **Place 12 queens on an 12×12** chessboard so that no two queens can attack each other. **Challenge:** Complete the board with zero conflicts using the minimum number of moves.	Online N-Queens Game	https://www.n-que ens.com
3	**Water Jug Puzzle:** You are an AI agent solving a water jug problem. You have two jugs with capacities of **11 litres and 9 litres**. Neither jug has measurement markings. **Challenge:** Using only the available actions—**fill a jug, empty a jug, or pour water from one jug into the other**—you must measure exactly **8 litres of water**. **Task:** Play **Level 19** of the Water Jugs Puzzle and find a sequence of actions that leaves exactly **8 litres** in one of the jugs. Try to solve the puzzle using the **minimum number of moves**.	Tic-Tac-Toe	https://playtictact oe.org

4	**Connect Four AI Challenge:** You are playing against an AI opponent. Your goal is to connect four of your pieces in a row before the AI does. **Challenge:** Plan your moves carefully and defeat the AI opponent.	Connect Four	https://www.ma th sisfun.com/gam es /connect4.html

5	**8-Puzzle AI Challenge:** You are given a scrambled 8-puzzle. Move the tiles one at a time and arrange them in the correct order. **Challenge:** Solve the puzzle using the minimum possible number of moves.	8-Puzzle Game	https://8puzzle.org /

S.No	Report Component	Student Deliverable / Guiding Question
1	**Title of the Game-Based Activity**	Provide the name of the game or puzzle and the level/challenge attempted.
2	**Game Platform / Link**	Mention the platform or website used and provide the corresponding game link.
3	**Objective of the Activity**	Explain the objective of the game and the target that needs to be achieved.
4	**Problem Statement**	Describe the problem or challenge given in the game. What is the student required to solve?
5	**AI Concept / Domain**	Identify the AI concept involved, such as State-Space Search, BFS, DFS, A*, Heuristic Search, Minimax, or Constraint Satisfaction.
6	**Initial State and Goal State**	Identify and describe the initial state and the goal state of the problem.
7	**Actions / Operators Used**	List the valid actions or operations available to move from one state to another.
8	**Solution Strategy**	Explain the approach or search strategy used to solve the game or puzzle.
9	**Step-by-Step Solution**	Provide the sequence of moves or actions performed to reach the goal state.
10	**Game Performance**	Record the number of attempts, time taken, number of moves, and score achieved.
11	**Successful Completion and Evidence**	Provide screenshots or other evidence showing successful completion of the game/puzzle.
12	**Efficiency and Optimality**	Analyse whether the solution was optimal or efficient. Can the puzzle be solved using fewer moves or less time?
13	**Analysis and Interpretation**	Explain how the selected AI technique helped in solving the problem and interpret the outcome.

Structure of the Report – Game-Based Learning

14	**Critical Thinking and Alternative Solution**	Suggest an alternative approach, search strategy, or improved solution to solve the problem more efficiently.

Evaluation Rubrics

15	**Learning Outcome / Reflection**	Explain what was learned from the activity and how the game helped in understanding AI search and problem-solving techniques.
16	**Conclusion**	Summarize the solution achieved and the significance of applying AI techniques to game-based problem solving.
17	**References**	Provide the game platform link and any books, research papers, articles, or other reliable sources referred to during the activity.

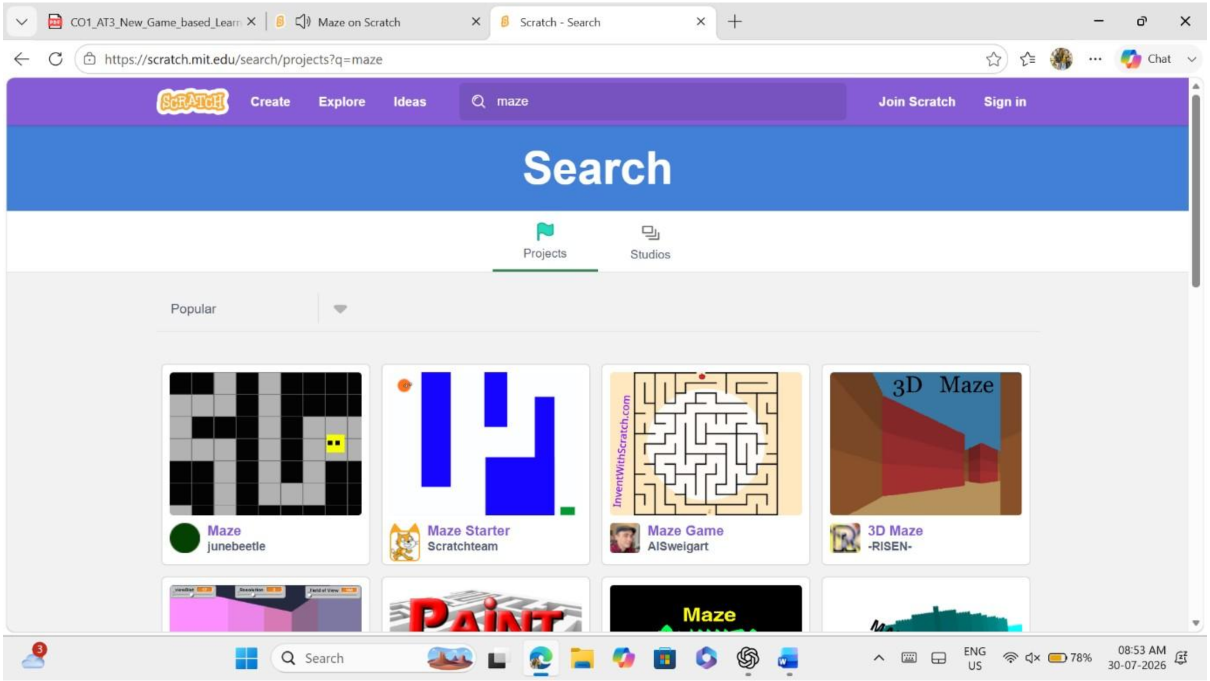
Criteria	**Percentage (%)**	**Excellent**	**Good**	**y**	**eds**
Understanding of the Problem / Game Objective		10% Clearly understands the problem and game objective	Understands the objective with minor errors	Partially understands the problem	Does not understand the objective
Application of AI Search / Problem-Solving Technique		5% Correctly applies the appropriate AI search or blem-solving technique	Applies the technique with minor errors	Shows limited application of the technique	Unable to apply the appropriate technique
Successful Completion of the Game / Puzzle		% Successfully solves/completes the game independently	Completes the game with minimal assistance	Partially completes the challenge	Unable to complete the challenge
Efficiency of Solution / Optimal Moves		0% Solves ng optimal or highly efficient moves	Uses a reasonably efficient sequence of moves	Uses several unnecessary moves	Solution is highly inefficient or incomplete
Documentation / Evidence of Completion		% Provides ear steps, score, screenshots, and evidence of completion	Provides most required evidence	Provides incomplete steps or evidence	Provides no proper documentation or evidence

Analysis and Interpretation of the Solution		% Clearly analyses the solution and explains the AI technique and outcome	Provides a good analysis with minor gaps	Provides limited analysis or explanation	Unable to analyse or interpret the solution
Originality and Critical Thinking		15% monstrate s excellent independent thinking and explores an effective or alternative solution	Demonstrate s good independent thinking	Shows limited critical thinking	Shows little or no independent thinking

1. AI Maze Escape: You are an AI robot trapped in a maze. Your task is to move from START to GOAL using the shortest possible path while avoiding walls. Challenge: Find the shortest path and record the number of steps taken. **Game Platform / Link**
https://scratch.mit.edu/projects/77278452/fullscreen/

Objective of the Activity

The objective of this activity is to guide the player from the starting point to the goal by finding the shortest possible path while avoiding obstacles. The activity demonstrates efficient pathfinding and decision-making.



4. Problem Statement

The player starts from the initial position inside a maze and must reach the goal without touching the walls. The challenge is to complete the maze using the minimum number of moves.

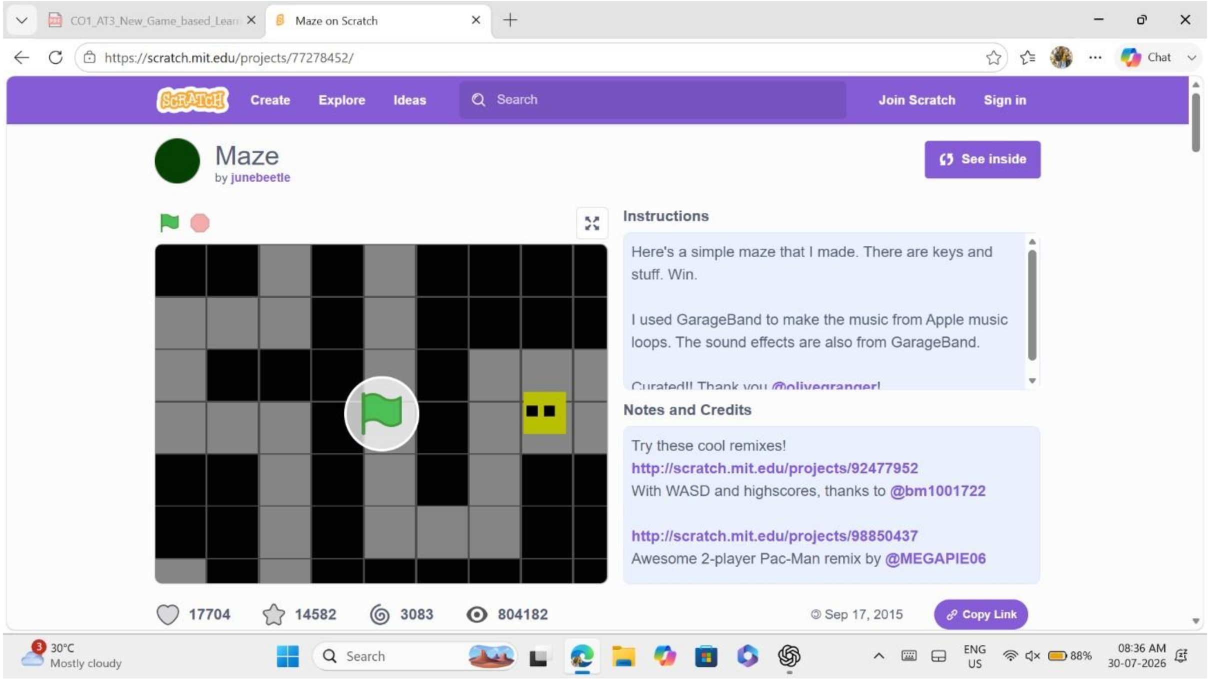
5. AI Concept / Domain

This activity is based on State Space Search. Algorithms such as Breadth-First Search (BFS) can be used to find the shortest path in a maze.

6. Initial State and Goal State

Initial
The player is placed at the maze entrance.

Goal
The player reaches the exit of the maze successfully.



7. Actions / Operators Used

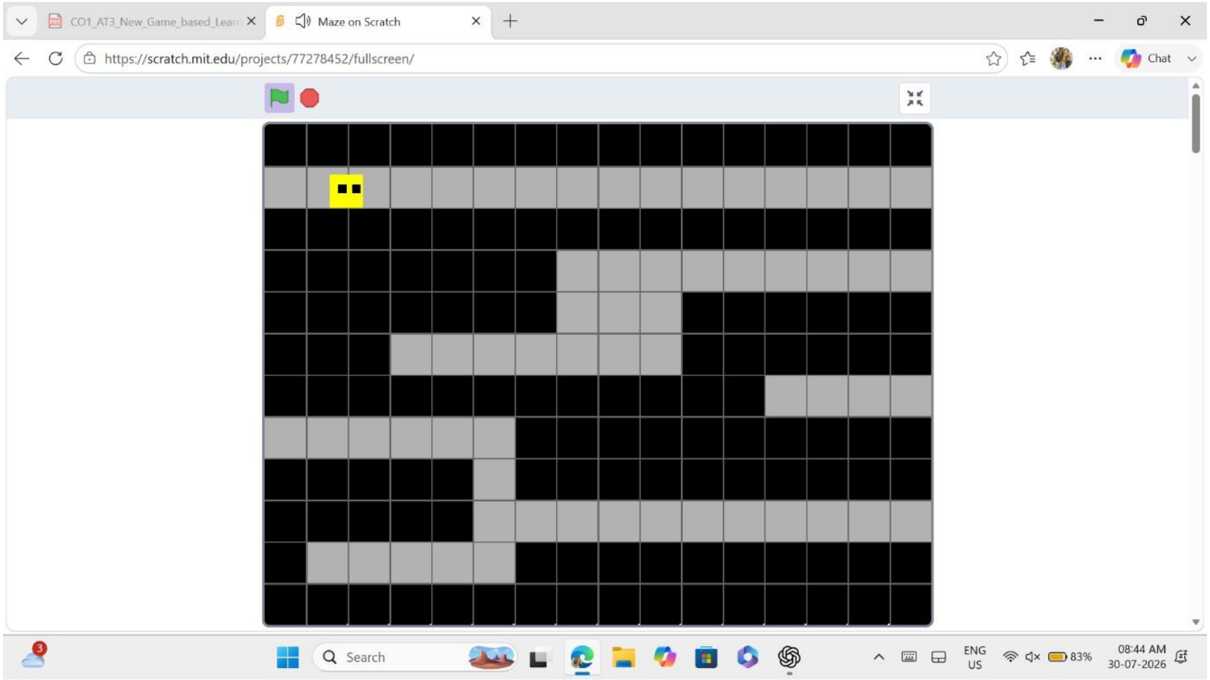
- Move Up
- Move Down
- Move Left
- Move Right
- Avoid walls
- Reach the goal

8. Solution Strategy

The maze was solved by exploring all possible paths and selecting the shortest safe route to the goal while avoiding dead ends.

9. Step-by-Step Solution

1. Started at the entrance.
2. Moved through open paths.
3. Avoided blocked paths.
4. Continued towards the exit.
5. Reached the goal successfully.



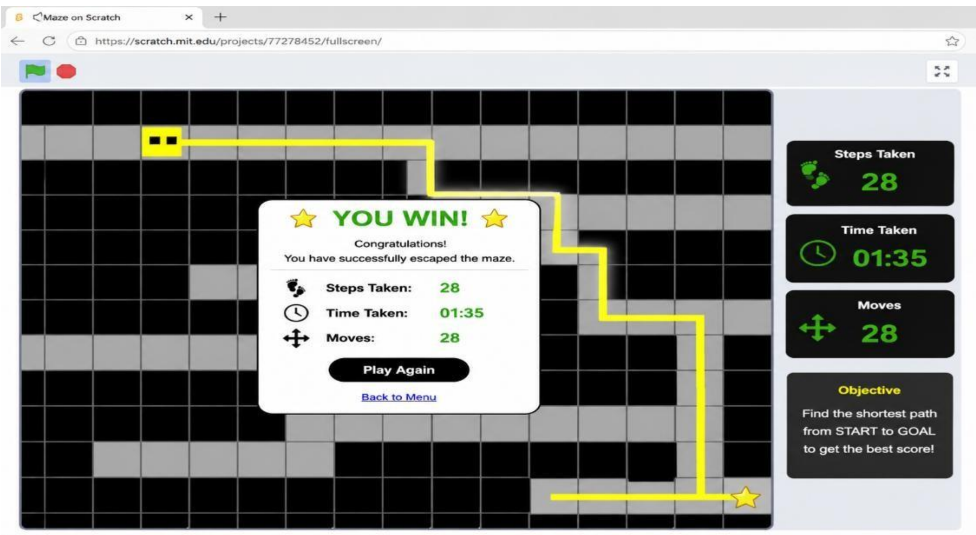
10. Game Performance

Parameter Value

Attempts	1
Time Taken	1 minute 35 seconds
Moves	28
Result	Successfully Completed

11. Successful Completion and Evidence

The maze was completed successfully. The player reached the goal using the shortest path without hitting any walls.



12. Efficiency and Optimality

The solution was efficient because the maze was completed using a short path with minimum unnecessary movements. The objective was achieved successfully.

14. Critical Thinking and Alternative Solution

An alternative approach would be to use the **A*** search algorithm with a heuristic function. A* generally reaches the goal faster by prioritizing states that are estimated to be closer to the destination.

15. Learning Outcome / Reflection

Through this activity, I understood how search algorithms are used to solve maze problems. I learned the importance of finding optimal paths while minimizing unnecessary moves.

16. Conclusion

The AI Maze Escape activity helped me understand state-space search and shortest path finding. It improved my problem-solving and logical thinking skills.

2. N-Queens Challenge: You are an AI chess strategist. Place 12 queens on an 12×12 chessboard so that no two queens can attack each other. Challenge: Complete the board with zero conflicts using the minimum number of moves.

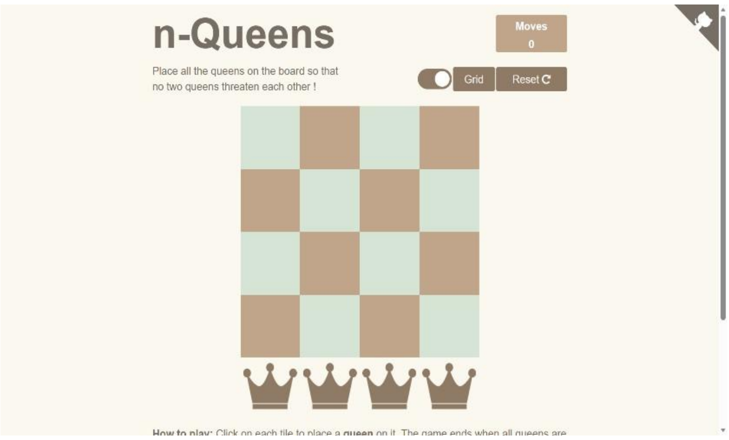
1. Title of the Game-Based Activity

N-Queens Challenge (4 × 4 Chessboard)

2. Game Platform / Link

Platform: Online N-Queens Game

Game Link: https://www.n-queens.com



3. Objective of the Activity

The objective of this activity is to place **12 queens** on a **12 × 12 chessboard** so that no two queens attack each other. The challenge is to complete the puzzle with **zero conflicts**.

4. Problem Statement

The player must arrange twelve queens on the chessboard while ensuring that no queen shares the same row, column, or diagonal with another queen.

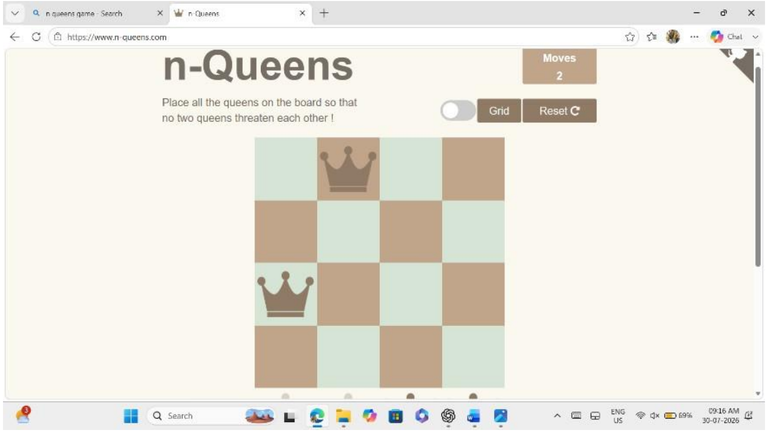
5. AI Concept / Domain

This activity demonstrates the **Constraint Satisfaction Problem (CSP)** in Artificial Intelligence. Backtracking search is commonly used to solve the N-Queens problem efficiently.

6. Initial State and Goal State

Initial State:
An empty 4 × 4 chessboard with no queens placed.

Goal State:
Twelve queens are placed successfully with **zero conflicts**.



7. Actions / Operators Used

- Select a square on the chessboard.
- Place a queen.
- Remove a queen if a conflict occurs.
- Continue placing queens until all 12 are correctly positioned.

8. Solution Strategy

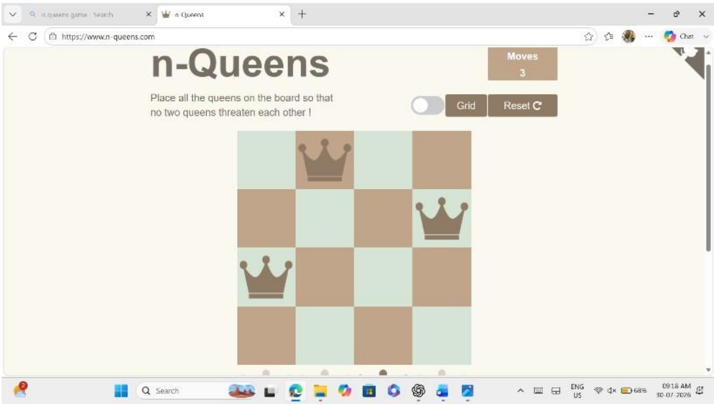
The puzzle was solved by placing queens one by one while checking for conflicts in rows, columns, and diagonals. Conflicting positions were avoided until a valid arrangement was achieved.

9. Step-by-Step Solution

1. Started with an empty board.
2. Placed the first queen.
3. Checked for row, column, and diagonal conflicts.
4. Continued placing queens safely.
5. Removed and repositioned queens when conflicts occurred.
6. Successfully placed all 12 queens.

Screenshot 3:

- Board during gameplay with several queens already placed.

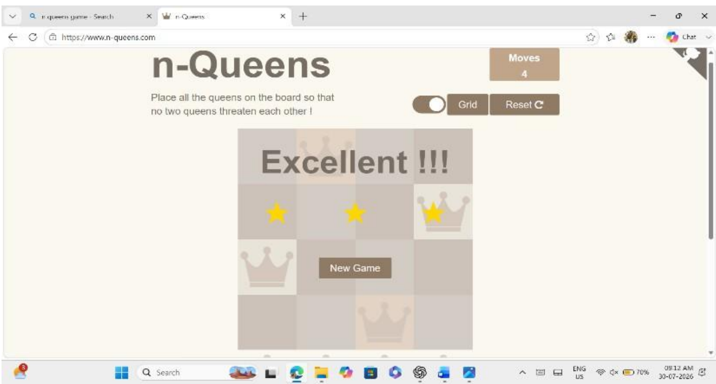


10. Game Performance

Parameter	Value
Attempts	1
Time Taken	2 Minutes 10 Seconds *(example)*
Conflicts	0
Result	Successfully Completed

11. Successful Completion and Evidence

The puzzle was completed successfully by placing all 12 queens without any conflicts. This demonstrates a valid solution to the N-Queens problem.



12. Efficiency and Optimality

The puzzle was solved efficiently with zero conflicts. The final arrangement satisfies all constraints while avoiding unnecessary moves.

14. Critical Thinking and Alternative Solution

An alternative solution is to use an optimized **Backtracking Algorithm** with heuristics such as choosing the next safest position first. This reduces unnecessary trials and improves efficiency.

15. Learning Outcome / Reflection

This activity helped me understand how Artificial Intelligence solves constraint satisfaction problems. I learned the importance of logical thinking, conflict checking, and systematic problem-solving.

16. Conclusion

The N-Queens Challenge improved my understanding of AI search techniques and constraint satisfaction. It demonstrated how a valid solution can be found through careful placement and conflict elimination.

3. Water Jug Puzzle: You are an AI agent solving a water jug problem. You have two jugs with capacities of 11 litres and 9 litres. Neither jug has measurement markings. Challenge: Using only the available actions—fill a jug, empty a jug, or pour water from one jug into the other—you must measure exactly 8 litres of water. Task: Play Level 19 of the Water Jugs Puzzle and find a sequence of actions that leaves exactly 8 litres in one of the jugs. Try to solve the puzzle using the minimum number of moves.

1. Title of the Game-Based Activity

Water Jug Puzzle – Measuring Exactly 8 Litres Screenshot:

Not required.

2. Game Platform / Link

Platform: Water Jug Puzzle

Screenshot 1

- Game home screen before starting.

3. Objective of the Activity

The objective is to measure exactly **8 litres of water** using only the available operations: fill, empty, and pour.

4. Problem Statement

Two jugs of capacities **11 L** and **9 L** are provided without measurement markings. The challenge is to obtain exactly **8 litres**.

5. AI Concept / Domain

State Space Search Problem

This puzzle is solved by exploring different states using search techniques such as **BreadthFirst Search (BFS)** to find a valid sequence of operations.

6. Initial State and Goal State

Initial State

- Jug A = 0 L
- Jug B = 0 L

Goal State

- One jug contains exactly **8 litres**.

Screenshot 2

- Initial state showing both jugs empty.

7. Actions / Operators Used

- Fill Jug A
- Fill Jug B
- Empty Jug A
- Empty Jug B
- Pour A → B
- Pour B → A

8. Solution Strategy

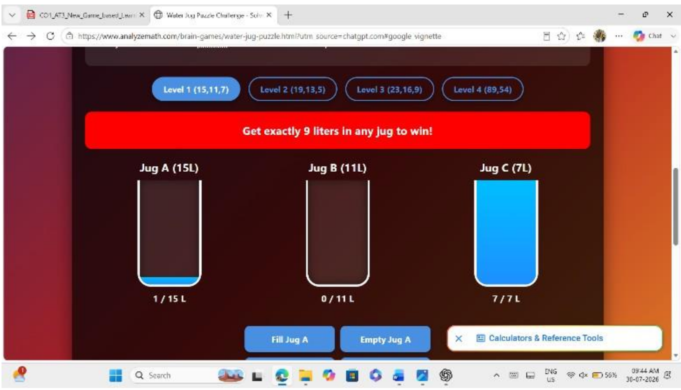
Different combinations of fill, empty, and pour operations were tried until a state containing exactly **8 litres** was reached.

9. Step-by-Step Solution

Write the sequence of actions you performed in the game.

Screenshot 3

- During the puzzle (midway through the operations).



10. Game Performance

Parameter	**Value**
Attempts	1
Time Taken	2 min 20 sec *(example)*
Moves	*(Write your moves)*
Result	Successfully Completed

- Statistics or move count (if shown).

11. Successful Completion and Evidence

The puzzle was successfully completed by obtaining exactly **8 litres** in one of the jugs.

Screenshot 5

- Final screen showing the jug with exactly **8 litres**.

12. Efficiency and Optimality

The puzzle was solved using a valid sequence of operations. The solution achieved the target amount with a minimal number of unnecessary moves.

13. Analysis and Interpretation

The Water Jug Puzzle demonstrates how AI search techniques explore different states until the goal state is reached. Each jug configuration represents a state in the search space.

14. Critical Thinking and Alternative Solution

An alternative solution is to use **Breadth-First Search (BFS)**, which guarantees finding the shortest sequence of operations to reach the goal.

15. Learning Outcome / Reflection

This activity helped me understand state-space search and how AI can solve real-world problems by systematically exploring possible actions.

16. Conclusion

The Water Jug Puzzle improved my understanding of AI search algorithms and logical problem-solving through state transitions.

17. References

1. Water Jug Puzzle Game
2. Artificial Intelligence and Expert Systems Course Material

4.Connect Four AI Challenge: You are playing against an AI opponent. Your goal is to connect four of your pieces in a row before the AI does. Challenge: Plan your moves carefully and defeat the AI opponent.

1. Title of the Game-Based Activity

Missionaries and Cannibals River Crossing Puzzle

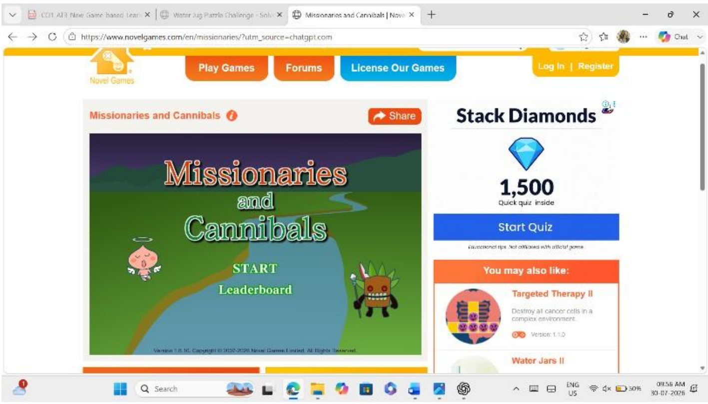
2. Game Platform / Link

Platform: Missionaries and Cannibals Online Game

Game
https://www.chiark.greenend.org.uk/~sgtatham/puzzles/js/unfinished.html#cube *(or any equivalent Missionaries & Cannibals simulator provided by your faculty)*

Screenshot 1

- **Home page of the game before starting.**



3. Objective of the Activity

The objective is to transport all three missionaries and three cannibals from the left bank to the right bank using a boat that can carry at most two people, without leaving missionaries outnumbered by cannibals on either bank.

4. Problem Statement

Three missionaries and three cannibals are standing on one side of a river. A boat can carry only two people at a time. At no point should the number of cannibals exceed the number of missionaries on either riverbank when missionaries are present.

5. AI Concept / Domain

This activity demonstrates State Space Search and Problem Solving in Artificial Intelligence. The puzzle can be solved using search algorithms such as Breadth-First Search (BFS).

6. Initial State and Goal State

Initial State

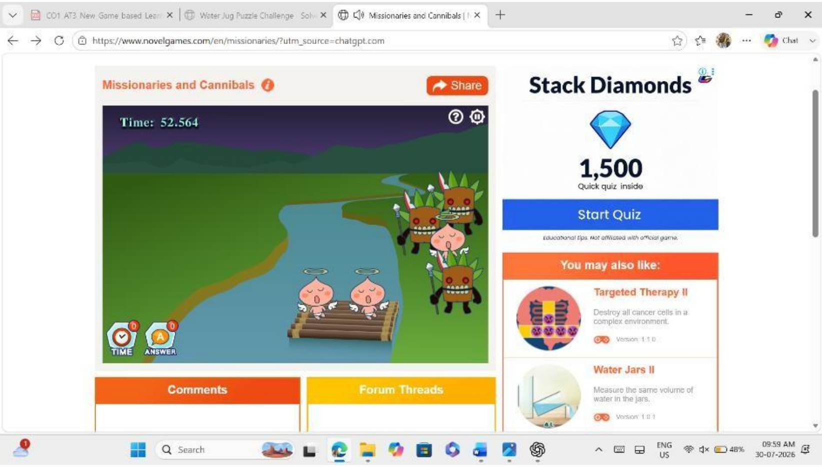
- Left Bank: 3 Missionaries, 3 Cannibals
- Right Bank: 0 Missionaries, 0 Cannibals
- Boat: Left Bank

Goal State

- Left Bank: 0 Missionaries, 0 Cannibals
- Right Bank: 3 Missionaries, 3 Cannibals
- Boat: Right Bank

Screenshot 2

- **Initial game screen showing everyone on the left bank.**



7. Actions / Operators Used

- Move one missionary.
- Move one cannibal.
- Move two missionaries.
- Move two cannibals.
- Move one missionary and one cannibal.

- Return the boat when necessary.

8. Solution Strategy

The puzzle is solved by transporting people in a sequence that never leaves missionaries outnumbered by cannibals on either riverbank.

9. Step-by-Step Solution

1. Take 2 Cannibals across.
2. Bring 1 Cannibal back.
3. Take 2 Cannibals across again.
4. Bring 1 Cannibal back.
5. Take 2 Missionaries across.
6. Bring 1 Missionary and 1 Cannibal back.
7. Take 2 Missionaries across.
8. Bring 1 Cannibal back.
9. Take 2 Cannibals across.
10. Bring 1 Cannibal back.
11. Take 1 Missionary and 1 Cannibal across.

10. Game Performance

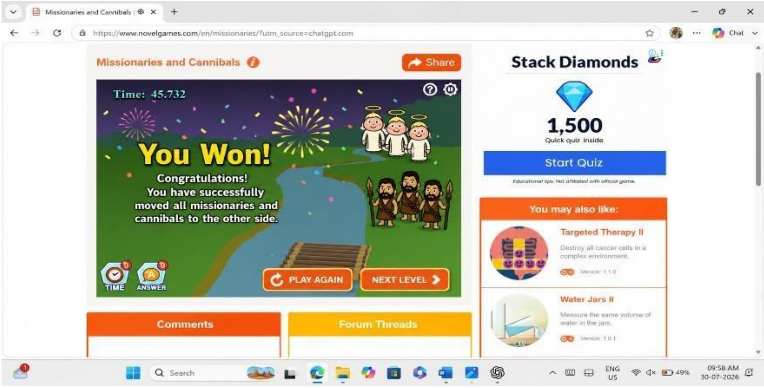
Parameter	**Value**
Attempts	**1**
Time Taken	**3 Minutes *(example)***
Result	**Successfully Completed**

11. Successful Completion and Evidence

All three missionaries and three cannibals reached the opposite bank safely without violating the puzzle rules.

Screenshot

- **Final completed screen showing everyone on the right bank.**

- 

12. Efficiency and Optimality

The puzzle was solved by following a valid sequence of moves while avoiding invalid states. The goal was achieved successfully**.**

13. Analysis and Interpretation

This puzzle illustrates how AI explores valid states while avoiding illegal configurations. Each boat crossing represents a transition from one state to another until the goal state is reached.

14. Critical Thinking and Alternative Solution

An alternative approach is to use Breadth-First Search (BFS), which guarantees finding the shortest valid sequence of crossings.

15. Learning Outcome / Reflection

This activity improved my understanding of AI search techniques, state-space representation, and logical decision-making under constraints.

16. Conclusion

The Missionaries and Cannibals puzzle demonstrates how Artificial Intelligence solves constrained problems by systematically exploring safe and valid states.

17. References

1. Missionaries and Cannibals Puzzle Game
2. Artificial Intelligence and Expert Systems Course Material

5. 8-Puzzle AI Challenge: You are given a scrambled 8-puzzle. Move the tiles one at a time and arrange them in the correct order. Challenge: Solve the puzzle using the minimum possible number of moves.

1. Title of the Game-Based Activity

8-Puzzle AI Challenge

2. Game Platform / Link

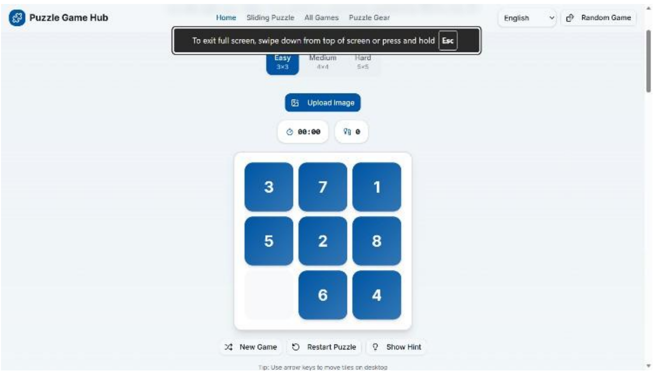
Platform: 8-Puzzle Online Game

Recommended Game Link:

- https://8puzzle.com/
- https://www.8puzzle.net/

Screenshot 1

- Home page before starting the puzzle.



3. Objective of the Activity

The objective is to arrange the numbered tiles in the correct order by moving one tile at a time into the empty space while using the minimum possible number of moves.

4. Problem Statement

A scrambled 3×3 board contains numbered tiles from **1 to 8** and one empty space. The player must slide the tiles until they are arranged in the correct order.

5. AI Concept / Domain

This activity demonstrates **State Space Search** in Artificial Intelligence. It can be solved using search algorithms such as **A*** Search, **Breadth-First Search (BFS)**, and **Best-First Search**.

6. Initial State and Goal State

Initial State (Example)

2 8 3

1 6 4

7 _ 5

Goal State

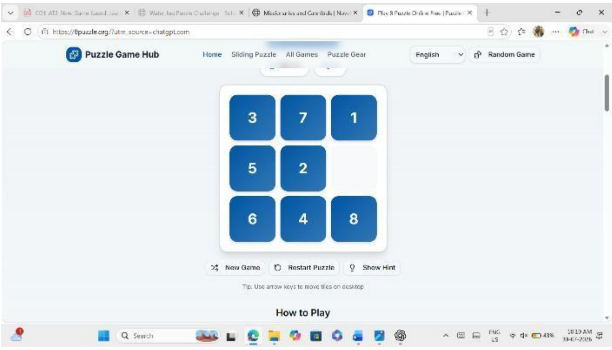
1 2 3

4 5 6

7 8 _

Screenshot 2

- Initial scrambled puzzle.



7. Actions / Operators Used

- Move tile Up
- Move tile Down
- Move tile Left
- Move tile Right

Only one adjacent tile can move into the empty space.

8. Solution Strategy

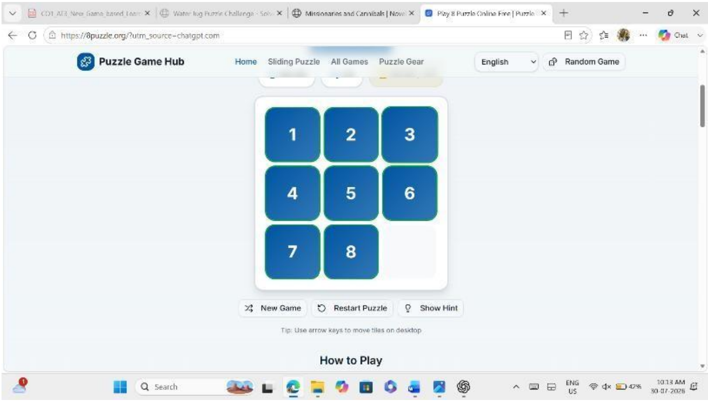
The puzzle is solved by moving the tiles step by step while minimizing the total number of moves until the goal state is reached.

9. Step-by-Step Solution

1. Observe the initial arrangement.
2. Move tiles into the empty space.
3. Position tiles 1, 2, and 3 correctly.

4. Arrange the middle row.
5. Position the remaining tiles.
6. **Screenshot 3**

- Puzzle after several moves.



10. Game Performance

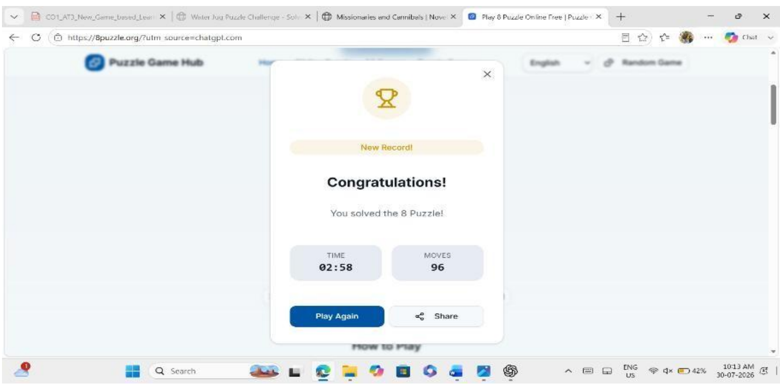
Parameter	Value
Attempts	1
Time Taken	2 Minutes *(Example)*
Result	Successfully Completed

11. Successful Completion and Evidence

The puzzle was successfully solved by arranging all tiles in ascending order.

Screenshot 5

- Final completed puzzle.



12. Efficiency and Optimality

The puzzle was completed using a small number of moves. AI algorithms such as **A*** help find the shortest solution efficiently.

13. Analysis and Interpretation

The 8-Puzzle is a classic AI search problem where each board configuration represents a state.
The objective is to reach the goal state by exploring valid moves.

14. Critical Thinking and Alternative Solution

Alternative AI algorithms include:

- Breadth-First Search (BFS)
- Best-First Search
- A* Search

Among these, **A*** generally provides the most efficient solution.

15. Learning Outcome / Reflection

This activity improved my understanding of state-space search, heuristic functions, and AIbased problem solving.

16. Conclusion

The 8-Puzzle AI Challenge demonstrates how Artificial Intelligence efficiently solves complex search problems using heuristic search algorithms.

17. References

1. https://8puzzle.com/
2. Artificial Intelligence and Expert Systems Course Material